OscTutorialPlugin

Manual and tutorial: nine ways to make a sawtooth

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1 Introduction

OscTutorialPlugin is a deliberately small synthesizer for teaching: one oscillator, one ADSR envelope, monophonic. Its point is the *oscillator model*: the same four waveforms (sine, triangle, rectangle, sawtooth) can be made in nine different ways, from the naive one that every beginner writes first, to band-limited tables, polynomial corrections and oversampling. You can switch between them while a note sounds, and you can *hear* the difference, *see* it on the built-in oscilloscope and spectrum and *measure* it (section 5.11).

(Until version 1.0.0 the plug-in was called VerySimpleSynth.)

Section 5 is a tutorial: why the naive oscillator aliases, how each method fights that, and what it costs. Section 6 has exercises for a lab session.

Warning: aliasing at high notes can be harsh and loud. Start with a low output level.

2 Installation

The downloaded file is a zip file. It contains the plug-in (.vst3; on macOS also the AU, .component), the Standalone application, a ReadMeFirst.txt, the licence files and this manual. Copy the plug-in to the right place and let your DAW rescan its plug-ins.

```
Windows (VST3):  C:\Program Files\Common Files\VST3
Mac (VST3):      /Users/yourUSERNAME/Library/Audio/Plug-Ins/VST3
Mac (AU):        /Users/yourUSERNAME/Library/Audio/Plug-Ins/Components
Linux (VST3):    /home/yourUSERNAME/.vst3
```

The Standalone application needs no DAW: choose the audio and MIDI devices in its Options menu and play with the on-screen keyboard or a MIDI keyboard.

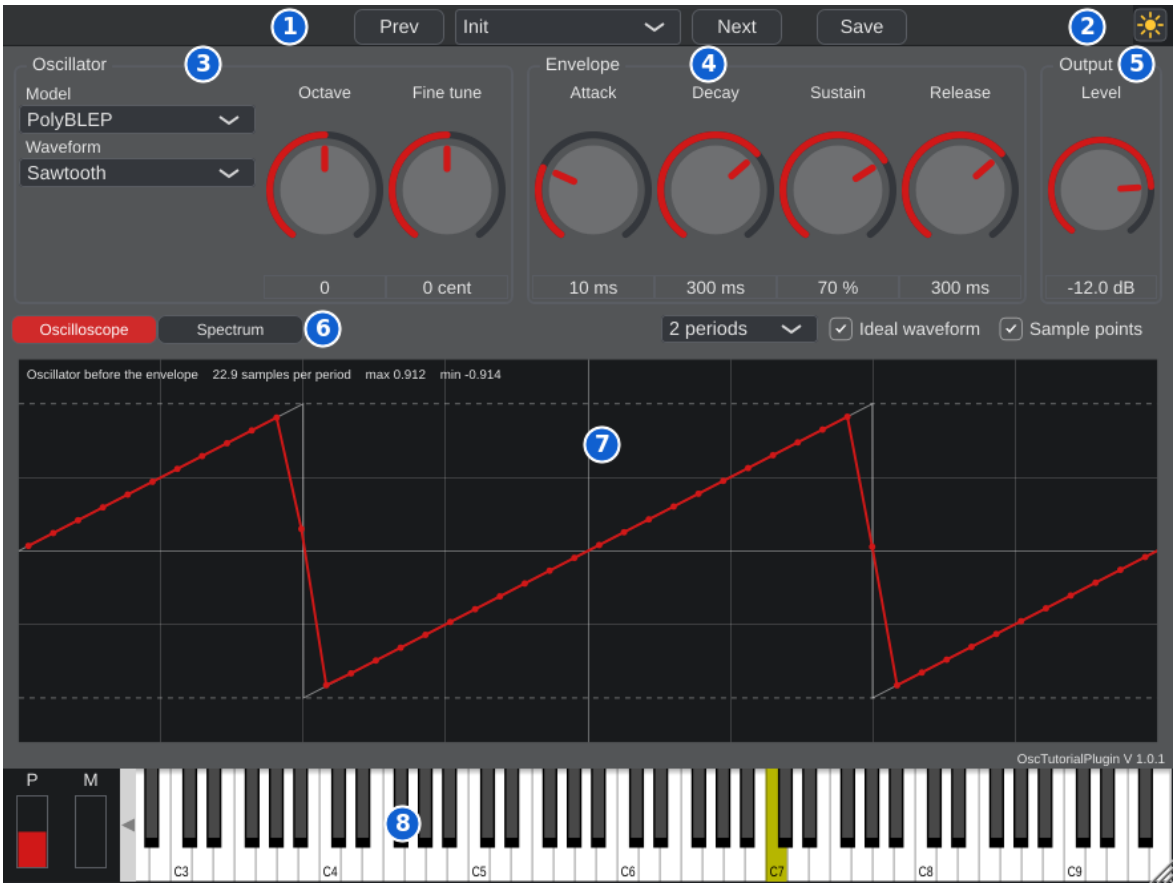
The macOS version is not signed with an Apple Developer ID. If macOS reports that the plug-in is damaged or cannot be opened, remove the quarantine flag in the Terminal, e.g. `xattr -cr ~/Library/Audio/Plug-Ins/VST3/OscTutorialPlugin.vst3`.

2.1 Presets and settings

The presets (plain XML files) and the settings file (`user.settings`: day/night theme) are stored in this folder – delete it to remove OscTutorialPlugin completely:

```
Windows: C:\Users\yourUSERNAME\AppData\Roaming\Jade_Hochschule\OscTutorialPlugin
Mac:     /Users/yourUSERNAME/Library/Audio/Presets/Jade_Hochschule/OscTutorialPlugin
Linux:   /home/yourUSERNAME/.config/Jade_Hochschule/OscTutorialPlugin
```

3 How to use



1. **Preset handler**: Prev/Next step through the presets. To save a new preset, type a name and press Save. Save turns red when you change a parameter of the displayed preset; press it to update the preset. To delete a preset, delete its XML file. The preset **Init** always holds the default values; it is written again at every start, so save your own sounds under another name.

2. **Day/night**: switches between the night (dark) and the day (light) theme.

3. **Oscillator**: *Model* (the method, section 5), *Waveform*, *Octave* ($-3 \ldots +3$) and *Fine tune* (± 100 cent).

4. **Envelope**: a linear ADSR (attack, decay, sustain level, release).

5. **Output**: the level (at velocity 127).

6. **Page buttons**: switch the big view between the oscilloscope and the spectrum.

7. **The big view.** *Oscilloscope*: 1 or 2 periods of the oscillator *before* the envelope (so the amplitude is always 1), started at the beginning of a period (phase 0), so the picture stands still. The thin line is the ideal waveform, the dots are the samples (shown when they are at least 4 pixels apart, i.e. for high notes). The top line shows the samples per period and the largest and smallest value. *Spectrum*: FFT of the oscillator (8192 points, Blackman-Harris window), 0 dB = a sine with amplitude 1, *linear* frequency axis up to $f_s/2$. The thin vertical lines mark the harmonics of the fundamental: every peak between them is aliasing. The top line shows the energy outside the harmonics (section 5.11).

8. **Keyboard** with pitch wheel (P, ±2 semitones) and modulation wheel (M, not used). The synth is monophonic: the last key pressed sounds; releasing it brings back the previous key that is still held.

4 Controls

All controls, their range and default value. Rest the mouse on a control to see its description as a tooltip.

Control	Range	Default	Description
Model	Naive - Oversampling 8x	Naive	How the oscillator makes the waveform: Naive (sampled ideal waveform, aliasing), Wavetable (band-limited tables, one per octave), Wavetable + Lanczos (harmonics weighted with Lanczos sigma factors: less overshoot at jumps), PolyBLEP and PolyBLEP 4-point (naive waveform with polynomial corrections at jumps and corners), minBLEP (minimum-phase corrections from a table), BLIT (integrated band-limited impulse trains), DPW (differentiated antiderivative), Oversampling 8x (naive at 8 fs, filtered).
Waveform	Sine - Sawtooth	Sawtooth	The waveform of the oscillator (Rectangle: 50 % duty cycle).
Octave	-3 - 3	0	Shifts the pitch by whole octaves.
Fine tune	-100 cent - 100 cent	0 cent	Fine tuning in cent (100 cent = 1 semitone).
Attack	1 ms - 5000 ms	10 ms	Time to rise to full level after a key is pressed.
Decay	1 ms - 5000 ms	300 ms	Time to fall from full level to the sustain level.
Sustain	0 % - 100 %	70 %	Level while the key is held (in % of full level).
Release	1 ms - 5000 ms	300 ms	Time to fall to silence after the key is released.
Level	-60.0 dB - 0.0 dB	-12.0 dB	Output level (at velocity 127).

5 Tutorial: the oscillator models

5.1 Why the naive oscillator aliases

A digital oscillator keeps a *phase* $p \in [0, 1)$ and adds $\Delta p = f_0/f_s$ at every sample (f_0: frequency of the note, f_s: sample rate). The simplest idea is to compute the ideal waveform at this phase: the sawtooth is $2p - 1$, the rectangle is ± 1.

The ideal sawtooth has a jump in every period, and a jump has infinitely many harmonics:

$$s(t) = \frac{2}{\pi} \sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k} \sin(2\pi k f_0 t).$$

Sampling cannot represent frequencies above $f_s/2$. A harmonic at $kf_0 > f_s/2$ is not lost, it *folds back* (aliases) to

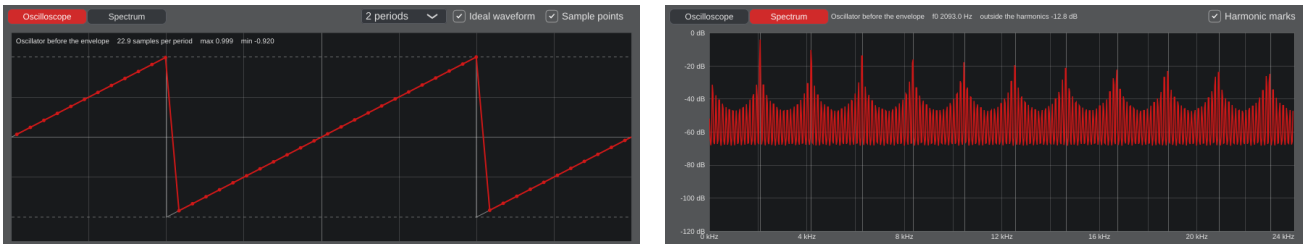
$$f_{\text{alias}} = |kf_0 - mf_s| \quad \text{(the integer } m \text{ that brings it below } f_s/2).$$

These aliases are not harmonics of f_0: they sound inharmonic, rough, like a ring modulator, and they even move in the opposite direction when you bend the pitch up. Their level falls only like $1/k$ (sawtooth, rectangle) or $1/k^2$ (triangle; it has corners, no jumps). So: the higher the note, the more harmonics lie above $f_s/2$, and the worse it gets. The sine has only one harmonic and never aliases (below $f_s/2$).

Every method below answers the same question differently: *how do we get the harmonics below $f_s/2$ without the ones above?*

5.2 Naive

Idea: the ideal waveform, evaluated at the phase of each sample.



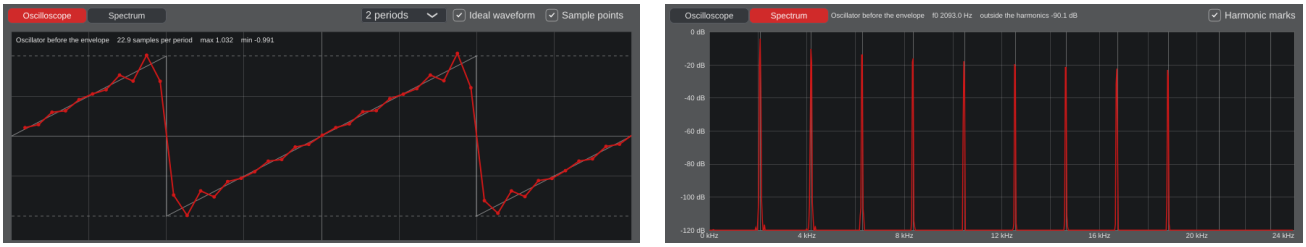
Naive: sawtooth at C7 (2093 Hz, 22.9 samples per period, 48 kHz). Left: oscilloscope, right: spectrum.

The oscilloscope shows the perfect shape – the jump falls exactly between two samples. That is the trap: the samples look right, but the spectrum is full of aliases (the dense lines between the harmonic marks; about −13 dB of the energy are aliases).

Pros: trivial, cheap, exact shape, no delay. **Cons:** strong aliasing for all waveforms with jumps or corners, getting worse with pitch. Acceptable only for low notes or for LFOs.

5.3 Wavetable

Idea: store one period of the waveform in a table, computed by additive synthesis with only the harmonics that fit below $f_s/2$. One table per octave of the fundamental (10 tables of 2048 samples per waveform; table k is used for fundamentals up to $40\,\text{Hz} \cdot 2^k$ and has all harmonics below $f_s/2$ for its highest note). Read with linear interpolation.



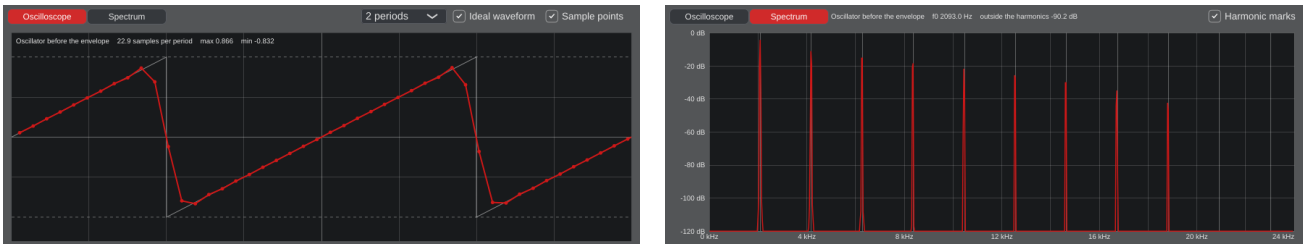
Wavetable: sawtooth at C7 (2093 Hz, 22.9 samples per period, 48 kHz). Left: oscilloscope, right: spectrum.

The spectrum is clean (−86 dB). The oscilloscope shows the price of a band-limited jump: it overshoots and rings (Gibbs phenomenon, see section 5.12). Note also that the spectrum ends at the 9th harmonic (18.8 kHz): at C7 the table for fundamentals up to 2560 Hz is used, and it can only have 9 harmonics. Notes in the lower part of an octave lose up to one octave of bandwidth.

Pros: the cheapest method of all when running (one table lookup and one interpolation), almost no aliasing, works for any waveform you can describe by its harmonics. **Cons:** memory (here 330 kB per model), tables must be built (here at `prepareToPlay`), brightness steps at the table boundaries (could be avoided by crossfading two tables), interpolation errors, and the waveform is fixed (no hard sync, no pulse-width modulation without extra tables).

5.4 Wavetable + Lanczos

The same tables, but harmonic k is weighted with the Lanczos sigma factor (section 5.12).



Wavetable + Lanczos: sawtooth at C7 (2093 Hz, 22.9 samples per period, 48 kHz). Left: oscilloscope, right: spectrum.

The overshoot is gone, the top harmonics are a little weaker (compare the spectra).

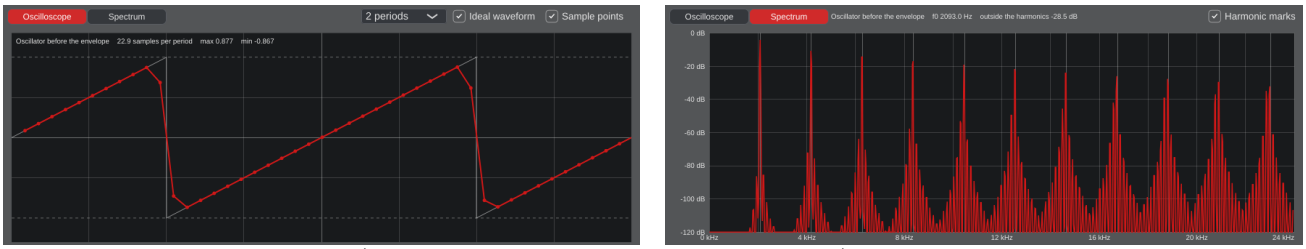
Pros: no ringing, otherwise as Wavetable. **Cons:** slightly duller, especially at high notes (few harmonics, so the sigma factors fall early).

5.5 PolyBLEP (2-point)

Idea: keep the naive waveform but repair each jump. A band-limited step (BLEP) differs from the ideal step only near the jump. PolyBLEP approximates this difference by a short polynomial and adds it to the samples around the jump. With d = distance to the jump in samples and a jump of height 1:

$$r(d) = \begin{cases} \frac{1}{2}(1+d)^2 & -1 < d \le 0 \text{ (before)} \\ -\frac{1}{2}(1-d)^2 & 0 < d < 1 \text{ (after)} \end{cases}$$

This is the integral of a triangular pulse of 2 samples width minus the ideal step. Corners (triangle) get its integral, the polyBLAMP $(1-|d|)^3/6$ per change of the slope. Because the phase tells us where the next jump is, the correction can also be applied *before* the jump.



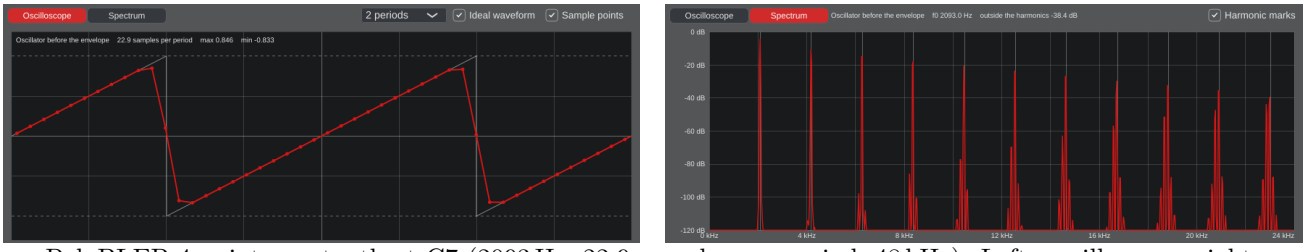
PolyBLEP: sawtooth at C7 (2093 Hz, 22.9 samples per period, 48 kHz). Left: oscilloscope, right: spectrum.

The dots show it: only the two samples next to each jump are changed. The spectrum is much better than naive (−29 dB), but the aliases near $f_s/2$ remain (the residual is short, so it is not a good low-pass).

Pros: cheap, no tables, no delay, works with changing frequency and with tricks like hard sync and pulse-width modulation (a jump can be anywhere), easy to understand. **Cons:** only moderate suppression; damps the top octave a bit.

5.6 PolyBLEP 4-point

The same with a smoother band-limited impulse: a cubic B-spline over 4 samples. The residual is a piecewise polynomial of degree 4 (degree 5 for the BLAMP), see `PolyBlepOscillator.cpp`.



PolyBLEP 4-point: sawtooth at C7 (2093 Hz, 22.9 samples per period, 48 kHz). Left: oscilloscope, right: spectrum.

About 10 dB better than the 2-point version (−38 dB), at almost the same cost. The B-spline is also a stronger low-pass: the harmonics near $f_s/2$ are weaker.

5.7 minBLEP

Idea: use a *long, good* band-limited step instead of a short polynomial: a windowed sinc (16 zero crossings on each side) integrated to a step. A symmetric step would need 16 samples of the future. The trick (Brandt 2001): make it *minimum phase* (via the real cepstrum), so all of its correction lies *after* the jump, and store it in a table (64 points per sample). At every jump the table is added to the next 32 samples.



minBLEP: sawtooth at C7 (2093 Hz, 22.9 samples per period, 48 kHz). Left: oscilloscope, right: spectrum.

The spectrum is as clean as the wavetable (−85 dB). The oscilloscope looks different: no ringing before the jump, more after it, and the whole waveform is **delayed** by about 3 samples. Next to the ideal curve, the rising ramp therefore lies *below* it – this is not an offset (the mean is 0), it is the delay: a ramp that comes 3 samples later is lower at the same moment.

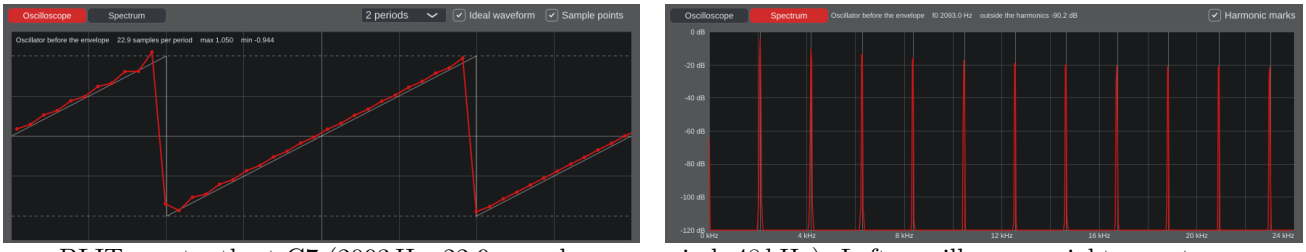
Pros: excellent suppression, works for jumps anywhere (hard sync, PWM). **Cons:** more complex (table, ring buffer of corrections), delay and changed phase of the harmonics, asymmetric overshoot (here up to 1.44 for the rectangle). Found while building it: with the cut-off of the sinc exactly at $f_s/2$, the transition band of the window lies half above $f_s/2$ and aliases; the cut-off is therefore $0.85\,f_s/2$.

5.8 BLIT

Idea (Stilson & Smith 1996): start from the derivative. The derivative of a sawtooth is a constant minus an impulse train; the derivative of a rectangle is a train of alternating impulses. A *band-limited* impulse train (BLIT) can be computed in closed form (Dirichlet kernel with $M = 2\lfloor P/2 \rfloor + 1$ terms, P = period in samples):

$$\mathrm{blit}(x) = \frac{1}{P}\,\frac{\sin(\pi M x)}{\sin(\pi x)}.$$

Integrating it gives band-limited waveforms. The integrators must be *leaky* (slightly below 1), otherwise numerical errors accumulate.



BLIT: sawtooth at C7 (2093 Hz, 22.9 samples per period, 48 kHz). Left: oscilloscope, right: spectrum.

Very clean (−83 dB). But the oscilloscope shows three side effects, all worth discussing: (1) the overshoot is larger than the Gibbs value (1.2 … 1.3): the running sum $y_n = y_{n-1} + x_n$ is not a true integrator, it boosts the harmonics near $f_s/2$ by up to $\pi/2$; (2) there is ringing near $f_s/2$ all over the period (the Dirichlet kernel decays slowly); (3) the leak (1 Hz) tilts the flat parts of the rectangle at low notes:



BLIT rectangle at C1 (32.7 Hz): the tilt of the leaky integrator and the ringing next to the jumps.

The triangle is a second integration of a rectangle. Two integrators in series amplify every small DC error enormously, so the triangle path needs a stronger leak (5 Hz).

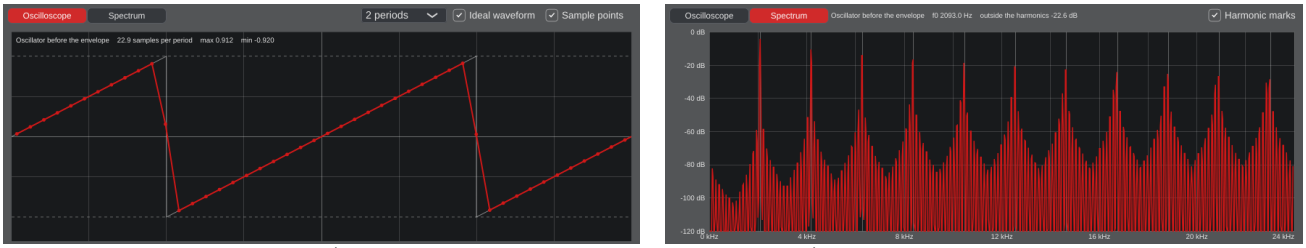
Pros: elegant theory, full bandwidth up to $f_s/2$, no tables. **Cons:** the most expensive of the analytic methods (several sin per sample), leaky integrators (tilt, start transients, DC), not easy for jumps at arbitrary times.

5.9 DPW (differentiated parabolic waveform)

Idea (Välimäki 2005): compute the *antiderivative* F of the waveform instead of the waveform itself and differentiate numerically:

$$y_n = \frac{F(p_n + \Delta p/2) - F(p_n - \Delta p/2)}{\Delta p}.$$

For the sawtooth $F = s^2/4$, a parabola – hence the name. This is the mean of the naive waveform over one sample interval: a box filter, so the harmonics fall like $1/k^2$ instead of $1/k$ before they are sampled.



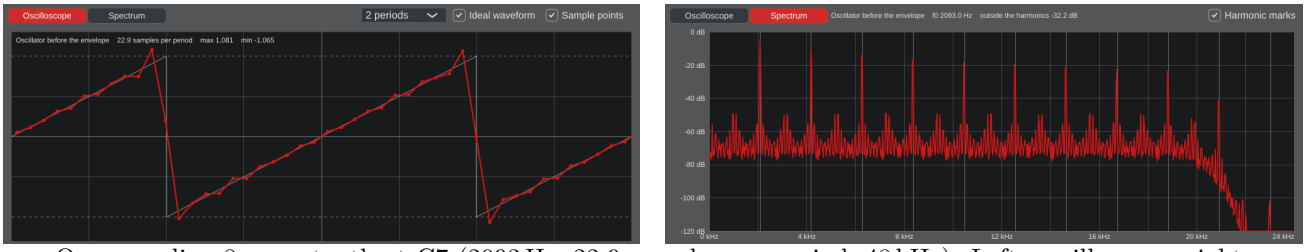
DPW: sawtooth at C7 (2093 Hz, 22.9 samples per period, 48 kHz). Left: oscilloscope, right: spectrum.

The jumps become ramps of one sample. About 10 dB better than naive (−23 dB).

Pros: extremely simple, cheap, no delay (centred difference), no tables, works for every waveform whose antiderivative you know. Higher orders (DPW3, DPW4: higher antiderivatives, repeated differences) are much better. **Cons:** first order is only a weak improvement; the difference of two large numbers loses precision at low notes (we use `double`).

5.10 Oversampling 8x

Idea: the brute-force method: run the naive oscillator at $8f_s$, low-pass filter (FIR, 512 taps, windowed sinc, cut-off $0.42f_s$) and keep every 8th sample. Only harmonics above $4f_s$ still fold back into the audio band, and they are weak.



Oversampling 8x: sawtooth at C7 (2093 Hz, 22.9 samples per period, 48 kHz). Left: oscilloscope, right: spectrum.

About $20 \log_{10} 8 = 18$ dB better than naive for the jumps (-32 dB); for the triangle (harmonics with $1/k^2$) much more (-80 dB). The filter rings like the wavetable (it is a band limit, too).

Pros: easy to understand, works for *everything*, also for non-linear processes (distortion, waveshaping, FM) where the other tricks fail. The general tool. **Cons:** by far the most expensive (8 naive samples and 512 multiplications per output sample), and for jumps still not very good; the filter delays (here compensated by running the oscillator ahead).

5.11 Comparison

Aliasing = energy outside the harmonics (and DC), relative to the total energy, in dB; the same number as on the spectrum page. Measured with `tools/analysis/measure_models.py` (1 s, Blackman window, 48 kHz). The floor of this measurement is about -57 dB at C4 and -93 dB at C7. **Cost** = computing time per sample (sawtooth, C7), measured on a desktop PC (Linux, gcc -O2); for comparison: 1 % of one core at 48 kHz is 208 ns per sample.

	sawtooth aliasing [dB]			rectangle	triangle	cost
Model	C4	C7	C8	max (C7)	C7 [dB]	[ns/sample]
Naive	−21.8	−12.8	−9.6	1.00	−40.3	5.6
Wavetable	floor	−85.5	−82.7	1.18	−92.5	3.7
Wavetable + Lanczos	floor	−88.4	−84.9	1.02	−92.8	3.7
PolyBLEP	−37.7	−28.5	−23.8	1.00	−54.0	11.6
PolyBLEP 4-point	−47.3	−38.4	−32.9	1.00	−65.0	13.5
minBLEP	floor	−85.3	−82.2	1.44	−92.5	19.4
BLIT	floor	−83.0	−79.9	1.30	−92.2	110
DPW	−31.9	−22.6	−18.5	1.00	−48.0	20.0
Oversampling 8x	−41.5	−32.2	−28.8	1.19	−80.2	396

Rules of thumb. Fixed waveforms and lowest cost: *wavetable* (with Lanczos if the ringing matters). Jumps at arbitrary moments (hard sync, PWM, changing shapes), no tables: *polyBLEP* (4-point if you can afford it), or *minBLEP* if quality matters and a few samples of delay do not. Anything non-linear: *oversampling*. *Naive* only for LFOs and low notes; *BLIT* and *DPW* mainly for understanding (and DPW of higher order as a cheap alternative to polyBLEP).

5.12 The Gibbs phenomenon and Lanczos sigma factors

A Fourier series that is cut off after H harmonics overshoots at every jump by about 9 % of the jump (for a rectangle from -1 to $+1$: up to 1.18), *however large H is*. More harmonics only make the ringing

narrower, not lower. Every band-limited jump rings like this – the wavetable, the oversampling filter, minBLEP and BLIT all show it.

Lanczos (1956) proposed to weight harmonic k with

$$\sigma_k = \operatorname{sinc}\left(\frac{k}{H+1}\right) = \frac{\sin\left(\pi k/(H+1)\right)}{\pi k/(H+1)}.$$

This is the same as averaging the cut-off series over one period of its highest harmonic: the ringing is smoothed away, the overshoot falls to about 1 %. The price is a gentle low-pass on the top octave. A sine is not a cut-off series and gets no sigma factor.



Rectangle at C5 (37 harmonics). Left: Wavetable, max 1.18. Right: Wavetable + Lanczos, max 1.02.

Try it with the presets *Gibbs Demo* and *Lanczos Demo*.

6 Exercises

1. **Hear aliasing.** Load *Aliasing Demo* (naive rectangle, two octaves up) and play a slow scale upwards. Describe what you hear. Bend the pitch with the wheel: in which direction do the aliases move?

2. **Predict an alias.** Naive sawtooth at C7 ($f_0 = 2093\,\text{Hz}$), $f_s = 48\,\text{kHz}$. Where does the 12th harmonic appear? (Answer: $12 f_0 = 25116\,\text{Hz} \to 48000 - 25116 = 22884\,\text{Hz}$.) Find it on the spectrum page.

3. **Compare the models.** On the spectrum page, play C7 with the sawtooth and step through all models. Note the value "outside the harmonics" and compare with the table above. Repeat with the triangle: why is the naive triangle so much better than the naive sawtooth?

4. **Where does the correction act?** Oscilloscope, PolyBLEP, C7: which samples differ from the ideal waveform? Now PolyBLEP 4-point and minBLEP.

5. **Gibbs.** Wavetable, rectangle: play C3, C5 and C7. What happens to the width and to the height of the ringing? Then Wavetable + Lanczos.

6. **Minimum phase.** minBLEP, sawtooth, C6: estimate the delay against the ideal curve. Why can a minimum-phase step not ring *before* the jump?

7. **BLIT.** BLIT, rectangle, C1 (Octave −1): explain the tilt. What would happen without the leak?

8. **Cost.** Which model would you choose for a 64-voice polyphonic synth on a small embedded processor? Which one for a distortion effect? Give reasons.

9. **Code.** Read `oscillators/PolyBlepOscillator.cpp` and derive the 2-point polyBLEP residual yourself (integrate a triangular pulse, subtract the step).

7 How it works

```
MIDI (DAW or on-screen keyboard)
  -> note stack (monophonic, last note), pitch bend, velocity
  -> oscillator (one of 9 models)  --> oscilloscope / spectrum
  -> linear ADSR envelope  -> level  -> both output channels
```

The frequency is $f = 440\,\mathrm{Hz} \cdot 2^{(n-69+12\,\mathrm{octave}+\mathrm{cent}/100+\mathrm{bend})/12}$ (n: MIDI note), limited to $0.45 f_s$. Parameters are read every 2 ms (synchronous blocks); notes start at the exact sample of the MIDI event. The internal block processing delays the output by 2 ms; the plug-in reports this latency to the host. All models are phase aligned (all waveforms start at 0 and rise), so the model can be switched while a note sounds; the new model continues at the phase of the old one.

8 Source code

OscTutorialPlugin is open source. You find the source code here: https://github.com/JoergBitzer/OscTutorialPlugin. Please report bugs there, too. Start reading with `oscillators/Oscillator.h`; each model has its own file. `planning.md` documents the decisions, the measurements and ideas how to go on.

9 References

- T. Stilson, J. Smith: Alias-free digital synthesis of classic analog waveforms. ICMC 1996 (BLIT).
- E. Brandt: Hard sync without aliasing. ICMC 2001 (minBLEP).
- V. Välimäki: Discrete-time synthesis of the sawtooth waveform with reduced aliasing. IEEE Signal Processing Letters 12(3), 2005 (DPW).
- V. Välimäki, A. Huovilainen: Antialiasing oscillators in subtractive synthesis. IEEE Signal Processing Magazine 24(2), 2007 (polyBLEP).
- V. Välimäki, J. Pekonen, J. Nam: Perceptually informed synthesis of bandlimited classical waveforms using integrated polynomial interpolation. JASA 131(1), 2012 (higher-order polyBLEP).
- F. Esqueda, V. Välimäki, S. Bilbao: Rounding corners with BLAMP. DAFx 2016 (polyBLAMP).
- C. Lanczos: Applied Analysis. Prentice Hall, 1956 (sigma factors).

10 Legal stuff

For this documentation the CC-BY 4.0 licence is valid. Copyright holder is Joerg Bitzer. The source code of the plug-in is under the MIT licence. The plug-in binaries contain JUCE (used under the AGPLv3), the VST3 SDK (MIT) and, on macOS, the Audio Unit SDK (Apache 2.0); the binaries as a whole are distributed under the AGPLv3.

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